

Social Media Analysis Project
The Devil Wears Prada 2
Network Structure, Sentiment and Dominant Narratives

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The Devil Wears Prada 2- Social Media Analysis project

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1. Objective and Research Design

1.1 Context and motivation

The Devil Wears Prada 2 is a useful case for social media analysis because it combines several overlapping cultural dimensions: cinema, fashion, celebrity culture, nostalgia, media discourse and box-office attention. These dimensions make the topic particularly suitable for studying how online conversations are structured around a cultural event and how different users attach different meanings to the same phenomenon.

Bluesky was selected as the platform for the analysis because it provides public textual conversations and reply structures that can be analysed both as content and as a social network. The project does not aim to measure the overall success of the film or to generalise to the entire audience. Instead, it focuses on how a specific platform discussed the phenomenon during a defined time window.

The analysis therefore investigates both what users talked about and how they interacted with each other. This dual perspective allows the project to combine textual analysis, sentiment and entity extraction with the study of reply networks and discussion communities.

1.2 Research question

How did Bluesky users engage with the release of *The Devil Wears Prada 2*, and how do discussion communities, sentiment and dominant themes differ across film-, fashion-, celebrity-culture- and box-office-related narratives?

1.3 Sub-questions

The research question is articulated into four sub-questions, which structure the analysis and are explicitly answered in the conclusion:

RQ1 — Network structure. What is the shape of the reply graph and what communities emerge?

RQ2 — Sentiment and emotion. Which affective tones dominate the discussion, and how do they vary across communities?

RQ3 — Temporal dynamics. How does the discussion evolve across the selected time window?

RQ4 — Entities and narratives. Which people, brands, works and cultural references dominate the conversation?

2. Data

2.1 Platform and data source

The dataset was collected from Bluesky using the AT Protocol through the atproto Python SDK. The collection includes public posts matching selected search queries and replies retrieved through thread expansion. The pipeline produces two main datasets: a post-level dataset and a reply-level dataset.

The post-level dataset contains the posts retrieved through search queries, while the reply-level dataset contains the replies retrieved from the expanded threads. This distinction is important because the posts are used to describe the overall conversation, while the replies make it possible to reconstruct the user-user interaction network.

2.2 Time window

The analysis covers the period from 1 April to 21 May 2026. The collection was performed on 21 May 2026, and the window was kept fixed after collection to ensure consistency and reproducibility across all notebooks.

2.3 Query strategy and pilot

A pilot collection was first used to test query productivity, topical relevance, author diversity and availability of replies. Based on the pilot results, the final collection refined the query set: queries that saturated the pilot cap were widened, low-signal queries were dropped, and six new queries were added to broaden recall, including sequel-related terms and informal references such as *dwp2*, *#dwp2*, *prada sequel* and *devil wears prada 2026*.

The final query set targets five overlapping audiences, reflecting the multi-dimensional nature of the topic.

Audience	Example queries
Film title and sequel	"the devil wears prada 2", "dwp2", "#dwp2", "prada sequel", "devil wears prada sequel", "devil wears prada" 2026, "prada 2" movie
Returning cast	"anne hathaway" prada, "meryl streep" prada, "emily blunt" prada, "stanley tucci" prada
Characters	"miranda priestly"
Fashion-industry referents	"anna wintour" prada
Press-tour guests	"lady gaga" prada, "doechii" prada

Table 1. Search queries used in the full collection, grouped by audience.

2.4 Dataset construction

The pipeline produces three derived datasets.

The posts dataset contains the search-collected posts and supports the corpus overview, the temporal analysis and the textual analysis.

The replies dataset contains replies retrieved through thread expansion, each linked to its parent author through the URI references, and is the source for the reply network.

Finally, posts and replies are merged into a unified text corpus of 6,158 unique texts authored by 2,407 distinct users, which serves as input for sentiment, emotion and Named Entity Recognition.

From the replies dataset, an edge list is derived in which each reply author is connected to the corresponding parent author. This list constitutes the input to the network analysis.

2.5 Data storage

Raw API outputs are stored in *data/raw*, derived analytical outputs in *data/processed*, and figures in *data/figures*. The pipeline is organised into four notebooks executed sequentially: pilot collection, full collection, network and community analysis, and content analysis.

3. Models and Methods

3.1 Reply network construction

The main network is a user-user reply graph. Nodes represent Bluesky authors, while edges represent reply interactions. If user A replies to content authored by user B, an edge is created from A to B. Therefore, the graph does not represent follower relations or general visibility, but direct conversational interaction.

Self-loops were removed because the unit of analysis is the interaction between different users. In Bluesky, self-replies often correspond to long thread-style posts split across multiple messages by the same author. These cases were flagged for interpretation but were not used as interaction edges.

Two graph versions were created. The directed graph preserves the direction of reply interactions and is used for in-degree and out-degree centrality. The undirected weighted graph is used for closeness, betweenness, eigenvector centrality and community detection.

3.2 Centrality measures

Five centrality measures were computed to describe the structural role of each author within the reply network. Each measure is interpreted as an interaction role rather than as a direct measure of popularity: popularity in social-media terms would require like and repost data, which are outside the scope of this reply-based analysis.

Measure	Computed on	Interpretation
In-degree	Directed graph	Authors receiving many replies — <i>attention targets</i>
Out-degree	Directed graph	Authors actively replying to others — <i>active commenters</i>
Betweenness	Undirected graph	Authors on many shortest paths — <i>bridges within the giant component</i>
Closeness	Undirected, giant only	Authors structurally close to others in the connected core
Eigenvector	Undirected graph	Authors connected to other central authors

Table 2. Centrality measures and their interpretation in the reply network.

3.3 Community detection

Community detection was applied to the giant component rather than to the full graph because the full network was highly fragmented. Applying community detection to all disconnected

components would mostly reproduce the existing component structure instead of identifying meaningful interaction communities.

Four algorithms representing distinct methodological families were compared. Their partitions were assessed using modularity, Adjusted Rand Index and Normalised Mutual Information. Louvain was selected as the reference partition because of its scalability, interpretability and alignment with the other modularity-based methods.

Algorithm	Family	Modularity	Role
Louvain	Modularity-based	0.420	Reference partition
Greedy Modularity	Agglomerative	0.420	Sensitivity check
Girvan–Newman	Divisive	0.411	Sensitivity check
Asynchronous FluidC	Propagation-based, fixed k	0.289	Alternative perspective

Table 3. Community detection algorithms compared on the giant component, with the modularity score of the partition each algorithm returned.

3.4 Text preprocessing

Posts and replies were combined into a single corpus. The text was standardised through lowercasing, removal of URLs, mentions, punctuation and special characters. English stopwords and project-specific collection keywords were removed, and tokens were lemmatised using WordNet.

Sentiment, emotion and NER were performed only on the English-tagged sub-corpus because the selected tools are English-oriented. Applying them to Portuguese, Spanish or French posts would risk generating artificially neutral or unreliable scores.

The original text was preserved for Named Entity Recognition because NER models rely on capitalisation, punctuation and contextual cues.

3.5 Sentiment and emotion analysis

VADER was selected as the main sentiment tool because it is designed for social media text. It produces a compound score that was converted into positive, neutral and negative labels using the standard thresholds of +0.05 and -0.05. AFINN was used as an additional polarity score to cross-check sentiment tendencies.

Emotion analysis was performed with NRClex, which extracts emotion categories such as anticipation, trust, joy, anger and sadness. This allows the analysis to move beyond positive and negative polarity and describe the affective structure of the conversation.

3.6 Named Entity Recognition

Named Entity Recognition was performed using SpaCy `en_core_web_sm`. The analysis focused on entity categories relevant to the research question: PERSON, ORG, GPE, LOC, WORK_OF_ART, EVENT and DATE.

NER was applied to the original text rather than to the aggressively cleaned version, because named entity models rely on capitalization, punctuation and contextual cues. The goal was to identify the people, works, organisations, places and events that anchor the conversation.

4. Results and Visualizations

4.1 Temporal dynamics of the conversation

The temporal distribution shows that the conversation is strongly event-driven. Daily volume is limited before mid-April and increases sharply after the beginning of May. Around 88% of the corpus belongs to the post-release period, suggesting that users reacted mainly to concrete events and visible media signals rather than maintaining a stable background discussion throughout the whole window.

This result is important because it contextualises the rest of the analysis. The network and sentiment patterns should be interpreted as the outcome of a concentrated release-window discussion, not as a long-term fandom conversation evenly distributed over time.

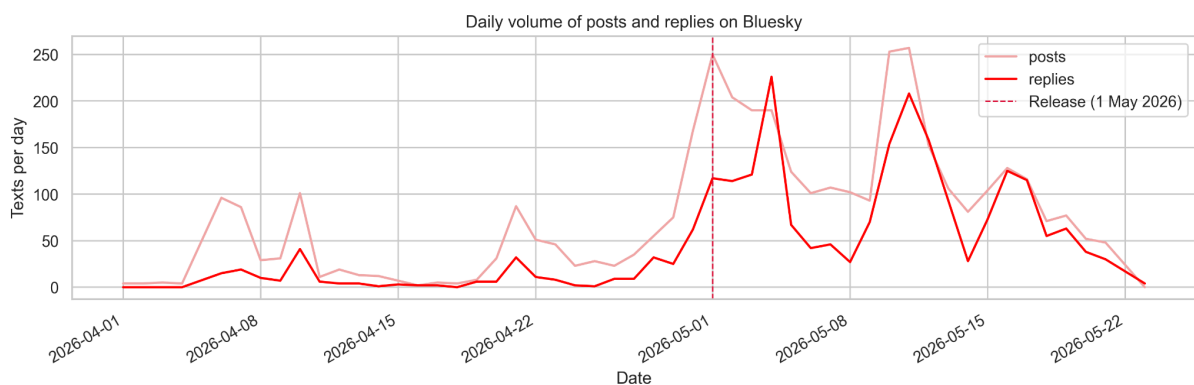


Figure 1. Daily volume of posts and replies in the analysed corpus. The conversation increases sharply around the release window, confirming the event-driven nature of the discussion.

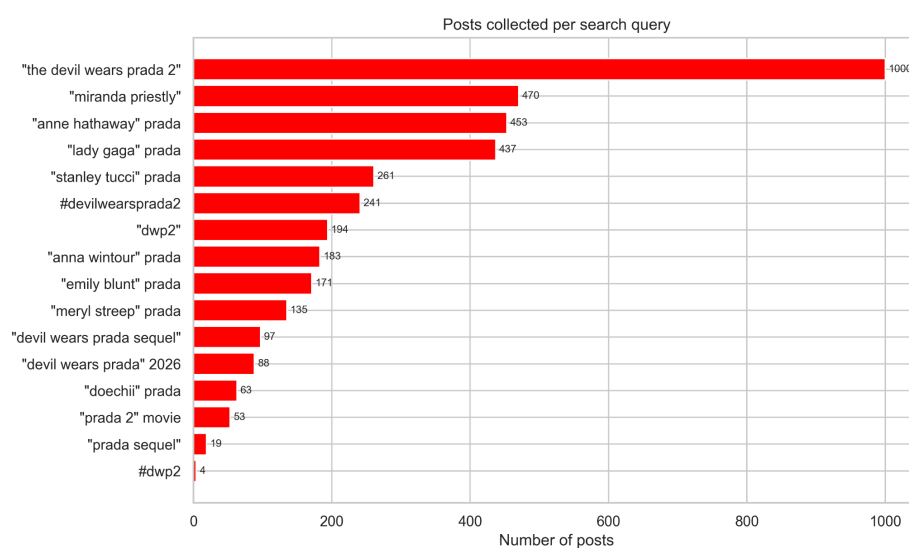


Figure 2. Query productivity in the full collection. The figure shows which search queries contributed most to the final corpus and helps evaluate the effectiveness of the query strategy.

4.2 General textual patterns

After preprocessing and removal of collection keywords, the most frequent lemmas provide a first overview of the recurring themes in the corpus. Beyond the search terms themselves, the ranking reveals four overlapping discussion axes that match the multi-dimensional structure of the research question.

First, *character-level discussion* emerges through terms such as *andy*, *character* and *original*, suggesting a recurring comparison between the new film and the 2006 original. Second, the *fashion dimension* is captured by *fashion* and *runway*, confirming that the fashion industry is central to the conversation. Third, the adjacent terms *box* and *office* (separated by tokenisation) signal a strong *box-office* framing. Fourth, evaluative terms such as *good*, *like* and *love* point to affective reactions that are then analysed in detail.

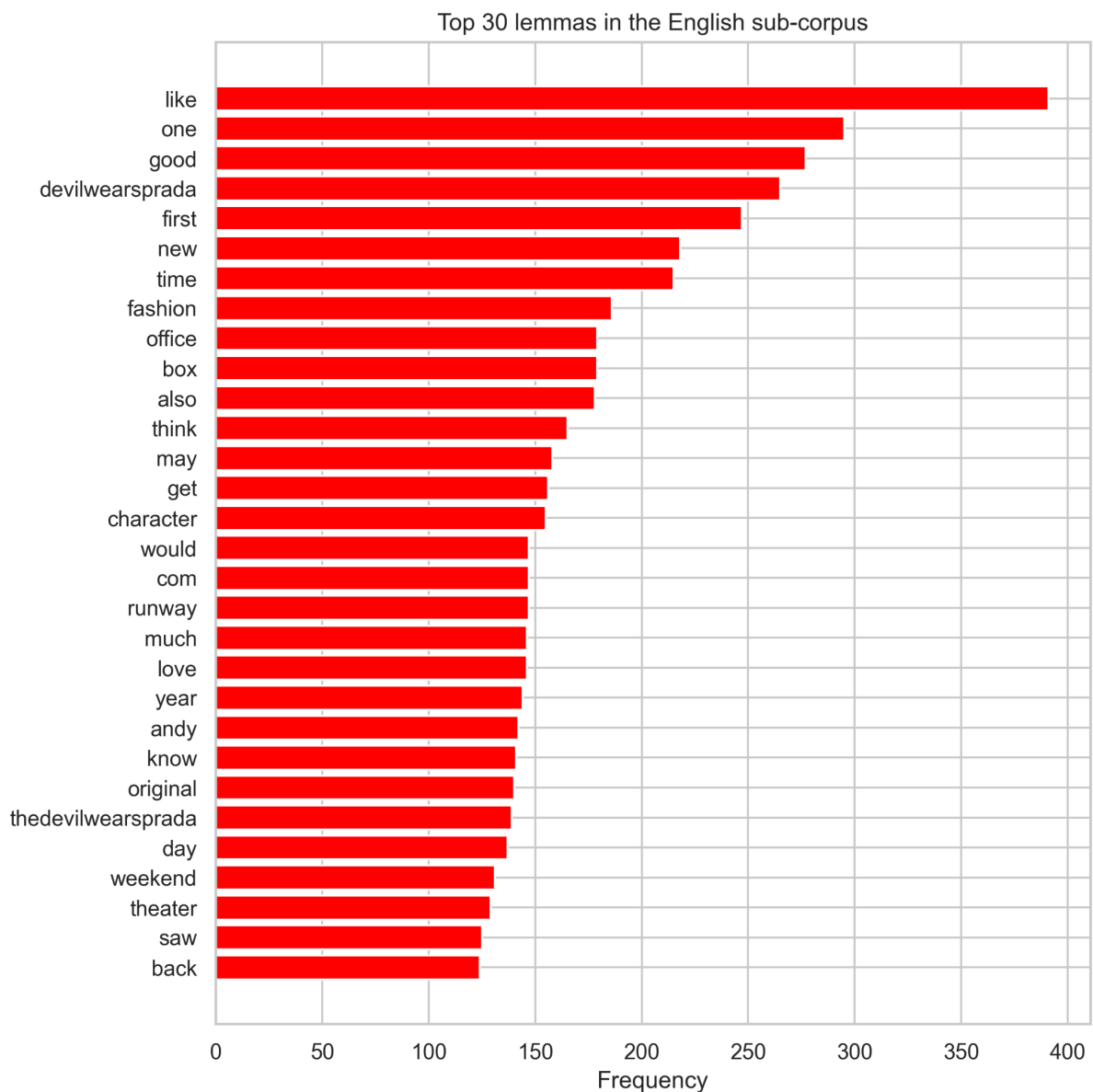


Figure 3. *Most frequent terms after preprocessing and removal of collection keywords. The figure highlights recurring discussion themes beyond the search terms themselves.*

4.3 Reply network structure

The reply graph reveals a highly fragmented conversation. The network contains 1,554 authors connected by 1,218 weighted edges, distributed across 371 disconnected components. The largest component includes only 5.7% of nodes (88 authors). This means that the Bluesky discussion did not develop as a single cohesive debate. Instead, it appears as an archipelago of small reply trees around individual posts.

This finding is central to the project because it changes the interpretation of the entire analysis. Since most users are not part of a single connected conversation, community-level results cannot be generalised to the whole corpus. The global text analysis describes the broader discourse, while the community analysis describes only the connected core of the network.

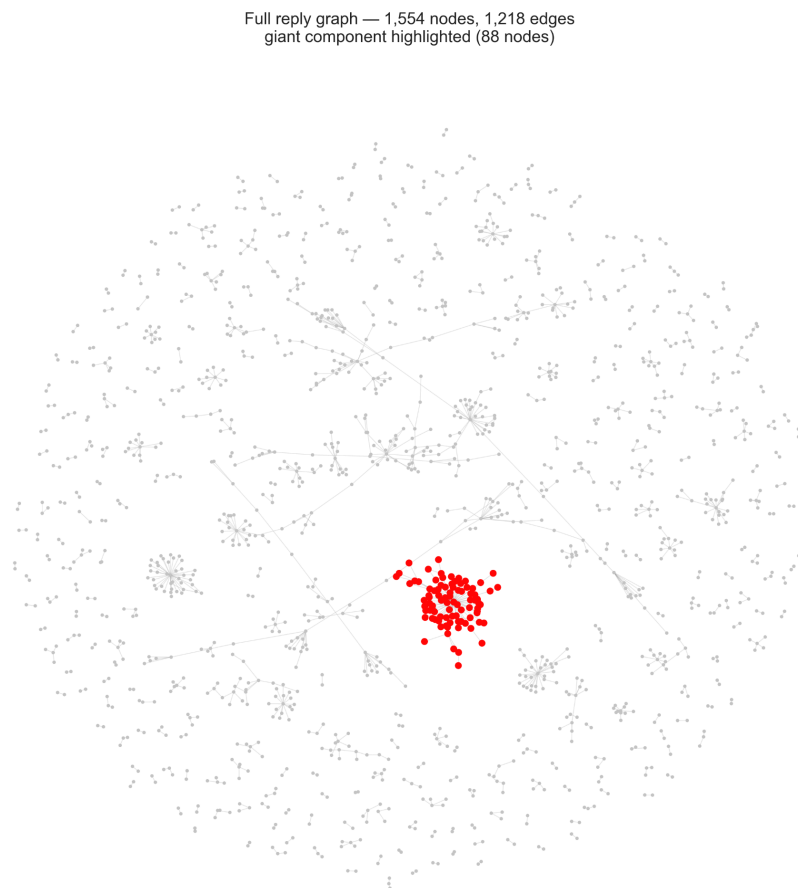


Figure 4. *Full Bluesky reply graph (1,555 nodes, 1,219 edges). The highlighted giant component is surrounded by many smaller disconnected components, showing that the discussion is fragmented into multiple micro-conversations rather than one cohesive debate.*

The visual fragmentation is confirmed quantitatively by the component-size distribution and by the degree distribution shown in the following figures.

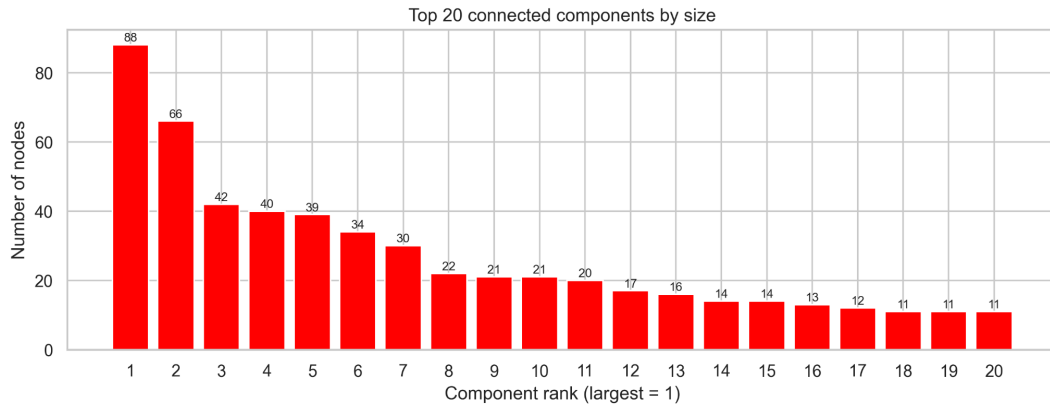


Figure 5. Size of the twenty largest connected components. The gradual decline after the giant component confirms that the conversation is distributed across several medium and small reply clusters.

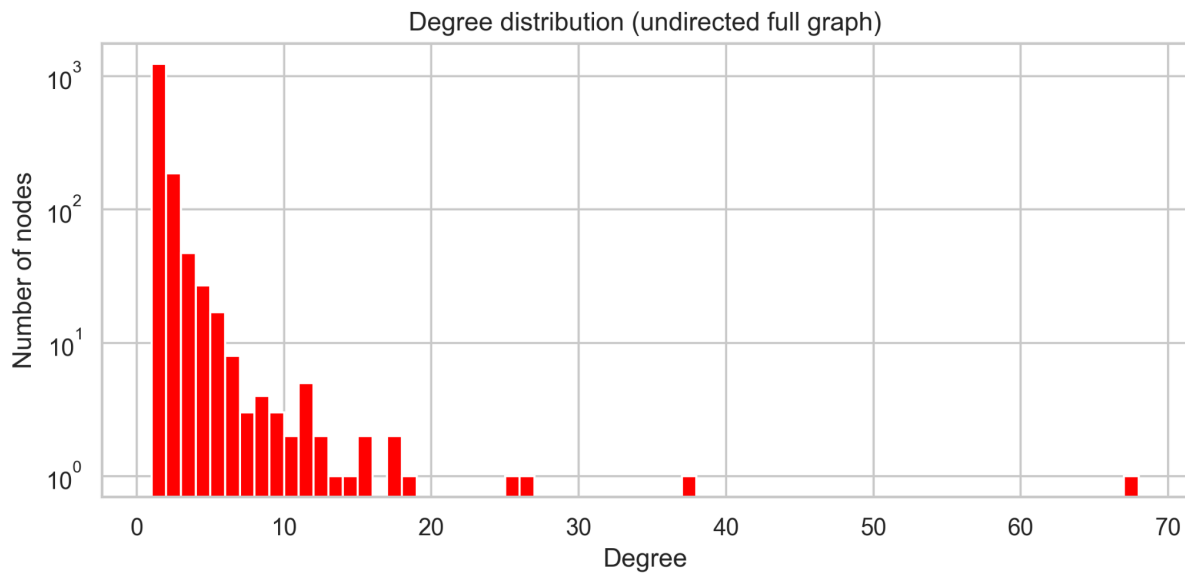


Figure 6. Degree distribution of the reply network on logarithmic scale. The distribution shows that most authors have few reply connections, while only a small number occupy more connected positions.

4.4 Central users and interaction roles

Centrality analysis shows that users play different interaction roles within the reply network. Some authors receive many replies and function as attention targets. Others have high out-degree and act as active commenters. Users with higher betweenness centrality can be interpreted as *bridge users within the giant component*: they connect distant nodes inside the

connected core, but not different connected components, which are disconnected by definition.

Centrality must therefore be interpreted carefully on this network. A user may be central within a specific component without being central to the entire Bluesky conversation. For this reason, centrality is treated as a measure of structural role inside the reply network, not as a direct measure of popularity or influence.

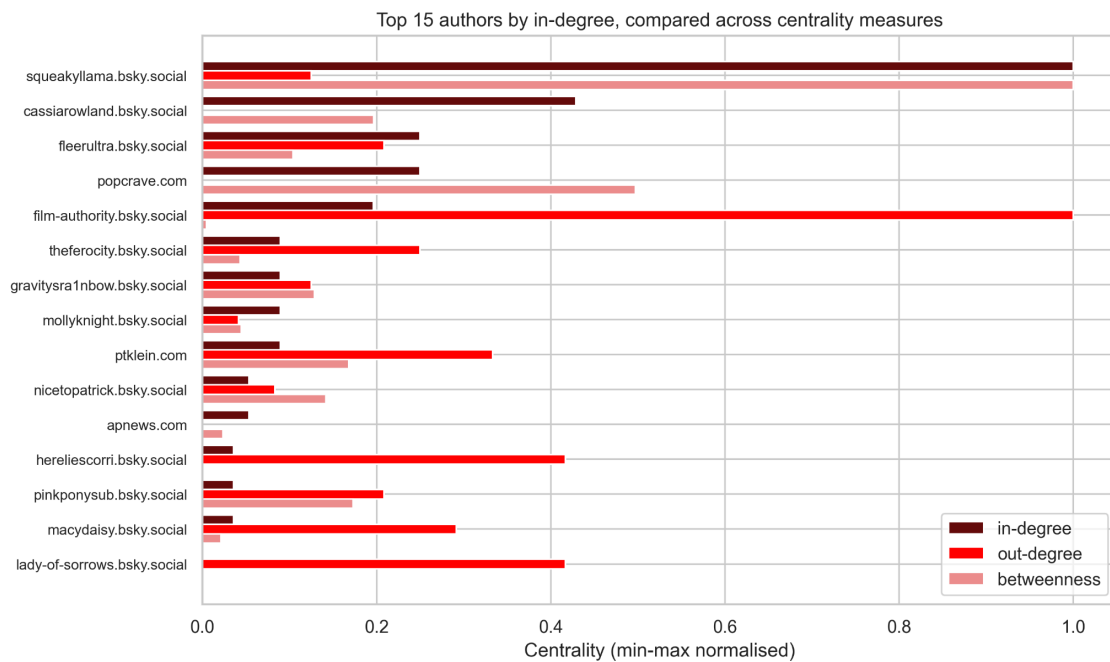


Figure 7. Comparison of normalised centrality scores for the most central authors. The figure distinguishes attention targets (high in-degree), active participants (high out-degree) and bridge users (high betweenness) within the reply network.

4.5 Community detection

The comparison between community detection algorithms shows that Louvain, Greedy Modularity and Girvan–Newman produce highly consistent partitions, with pairwise Adjusted Rand Index ≥ 0.96 and modularity values within a narrow band (0.411–0.420). FluidC returns a more divergent and coarser partition (modularity 0.289, ARI ≈ 0.38 against the other three), which is expected because it follows a diffusion-based logic and requires the number of communities to be specified in advance.

The convergence between Louvain, Greedy Modularity and Girvan–Newman (three algorithms of distinct methodological families) suggests that the giant component has a stable modular structure under different community-detection strategies. Louvain was therefore used as the reference partition for downstream content analysis.

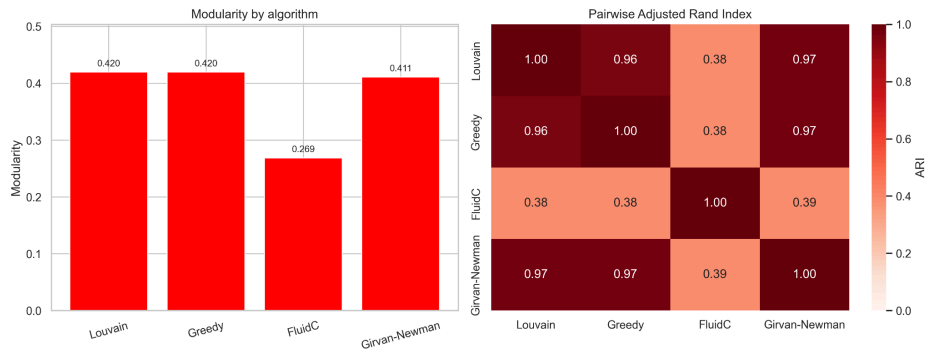


Figure 8. Community detection comparison. Louvain, Greedy Modularity and Girvan–Newman produce highly consistent partitions (pairwise $ARI \geq 0.96$), while FluidC returns a more divergent partition due to its diffusion-based logic and predefined number of communities.

The Louvain visualisation shows how the giant component is internally divided into interaction clusters. These clusters are then used to connect network structure with sentiment, emotions and entities.

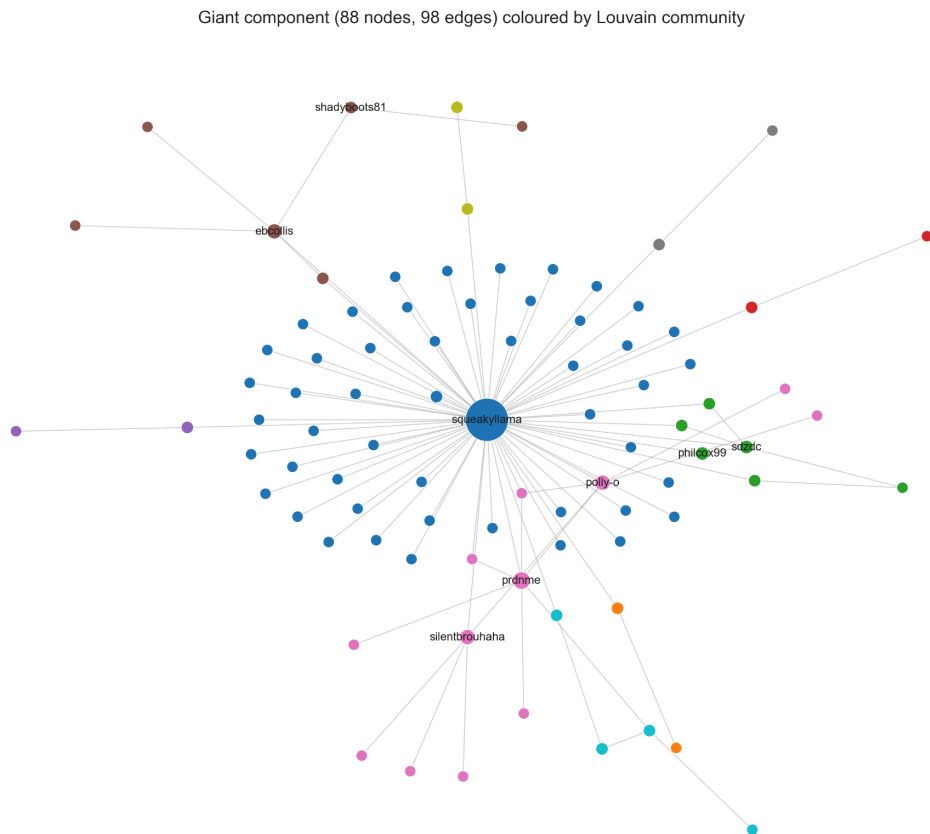


Figure 9. Giant component (88 nodes, 98 edges) coloured by Louvain community. The visualisation shows the modular structure of the main connected component and identifies the interaction clusters used for downstream content analysis.

4.6 Sentiment and emotion

Sentiment analysis shows that the discussion is polarised rather than uniformly positive. In the English sub-corpus, approximately 43% of texts are classified as negative, 38% as positive and 19% as neutral using VADER thresholds. This suggests that the release generated both enthusiasm and criticism.

The negative share should not be interpreted automatically as rejection of the film. In social media conversations, negative sentiment may reflect criticism, irony, disappointment, comparison with the original movie, or emotionally intense reactions to specific cast, fashion or box-office narratives. At the same time, the positive share indicates excitement, nostalgia and anticipation.

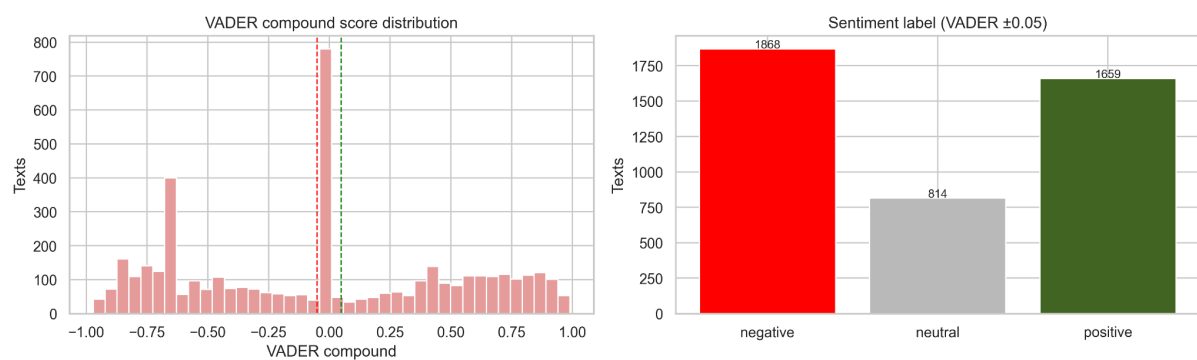


Figure 10. Sentiment distribution in the English sub-corpus. The discussion is polarized, with substantial shares of both negative and positive texts.

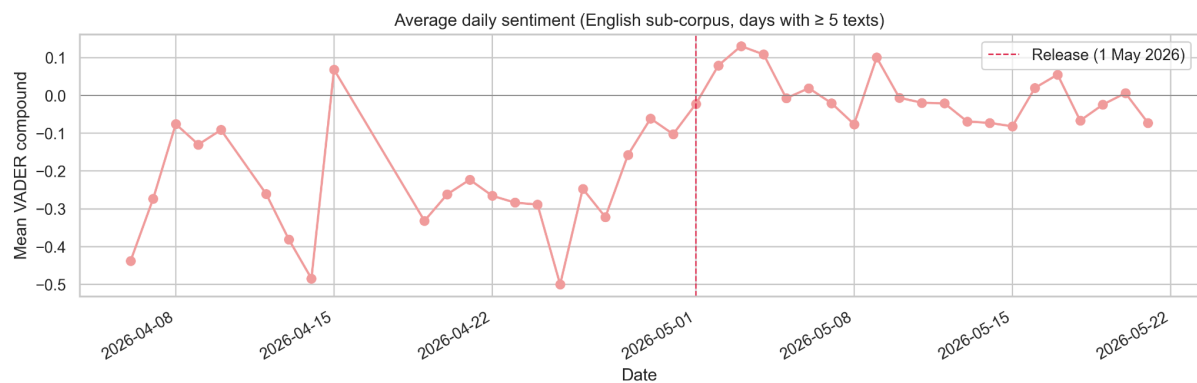


Figure 11. Average VADER sentiment over time. The figure shows how the tone of the conversation changes across the release window.

Community-level sentiment reveals differences across Louvain communities, but this interpretation must remain cautious because community labels exist only for the authors in the giant component. Therefore, this view is useful for understanding the connected core of the discussion, not the whole corpus.

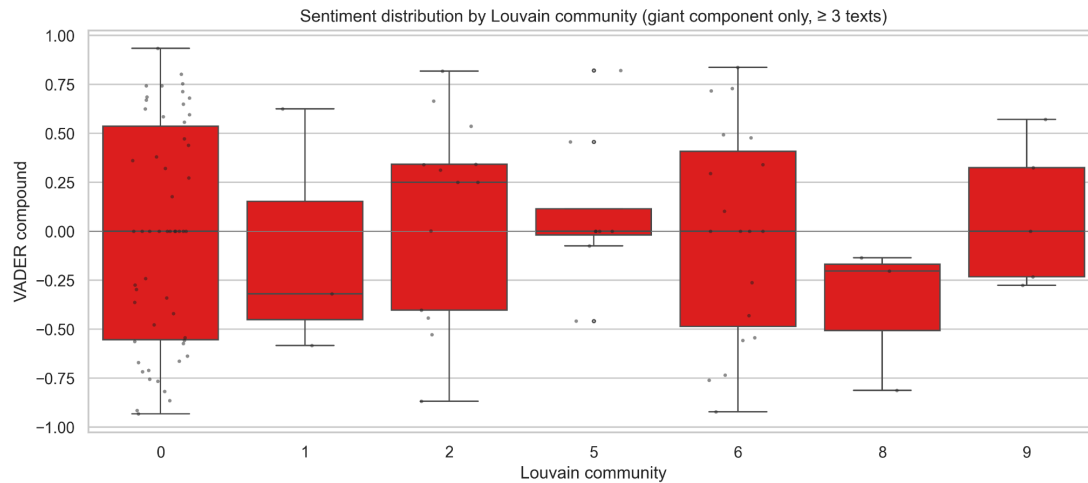


Figure 12. Sentiment distribution by Louvain community. Community-level sentiment is interpreted only for authors belonging to the giant component.

The emotion analysis complements sentiment by showing that the conversation is shaped by specific affective tones beyond polarity. The dominant emotions across the giant component are anticipation and trust, which are coherent with a release-window discussion around a familiar cultural franchise. The emotion heatmap highlights that different Louvain communities carry distinct affective profiles, with some clusters skewed towards *joy* and *surprise* and others towards sadness or disgust.

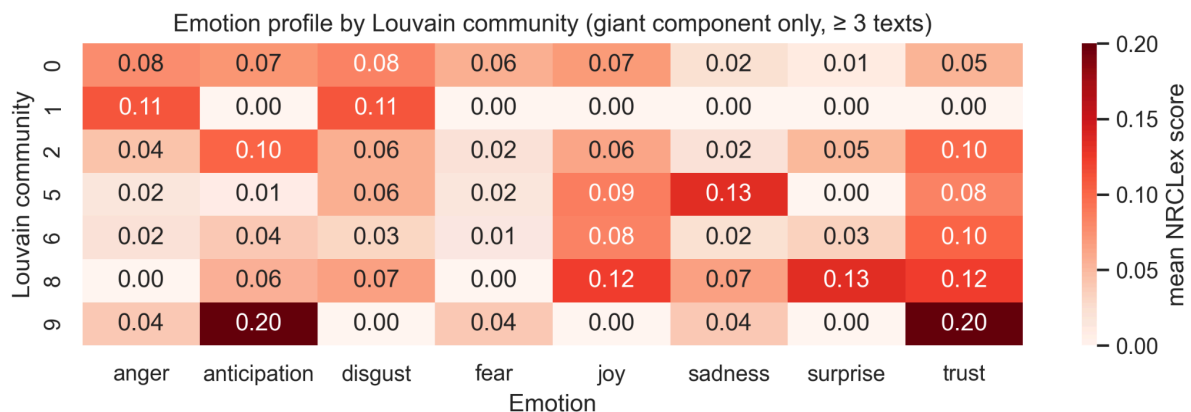


Figure 13. Emotion profile by Louvain community. The heatmap connects the network structure with affective patterns, highlighting differences in anticipation, trust, joy and negative emotions across communities.

4.7 Named entities and dominant narratives

Named Entity Recognition identifies the people, works, organisations and cultural references that anchor the conversation. The most frequent entities reveal that the discussion is not limited to the film as a standalone product, but extends to cast members (*Anne Hathaway*, *Meryl Streep*, *Emily Blunt*, *Stanley Tucci*), characters (*Miranda Priestly*, *Andy Sachs*),

fashion-industry referents (*Anna Wintour*, *runway*) and press-tour guests (*Lady Gaga*, *Doechii*).

The comparison between entities appearing in positive and negative texts helps interpret which names or references are associated with different affective tones. This does not mean that an entity is objectively positive or negative, but that it appears more often in texts with a certain sentiment orientation.

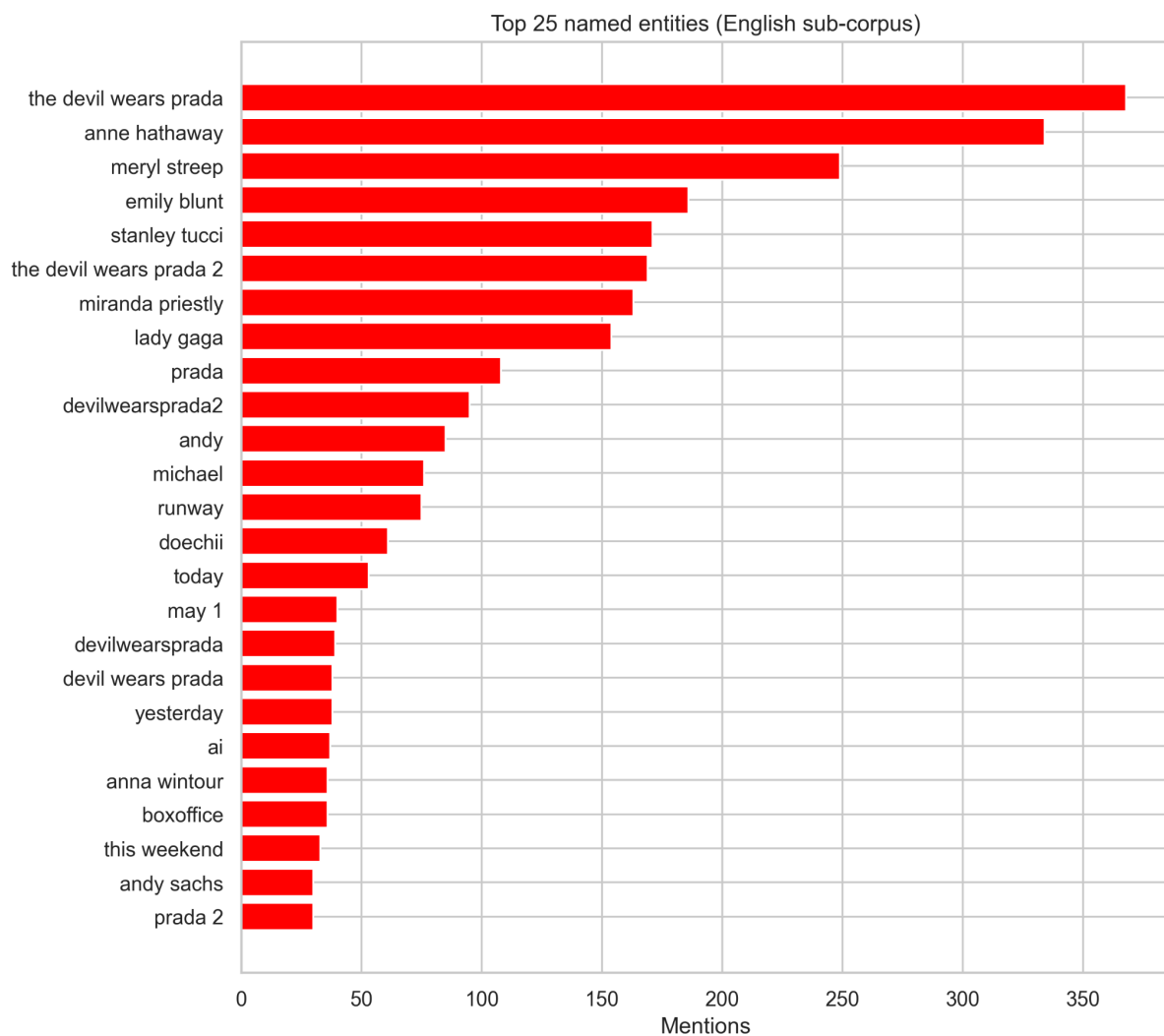


Figure 14. Most frequent named entities in the English sub-corpus. The results show which people, works, organizations and cultural references anchor the conversation.

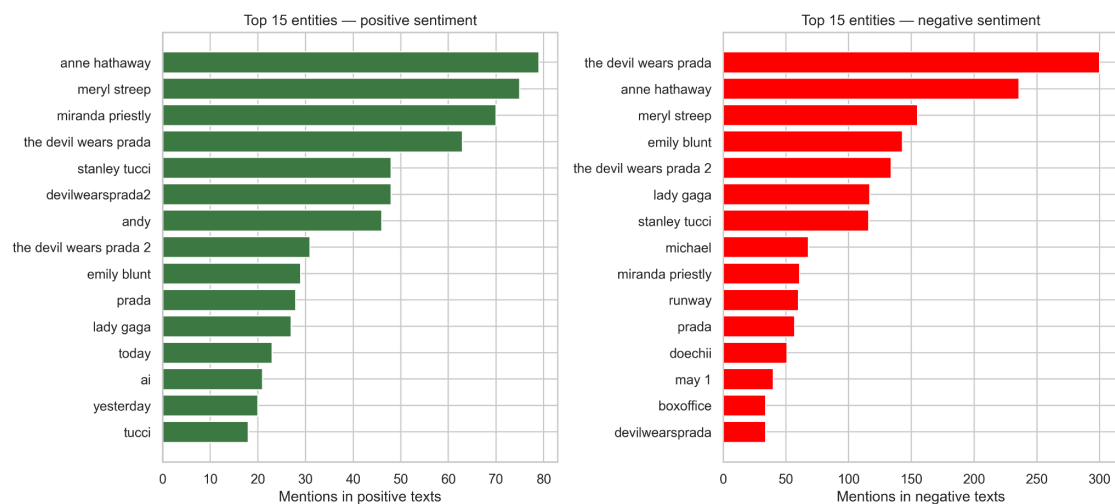


Figure 15. Named entities in positive and negative texts. The comparison highlights how different entities are associated with different affective tones.

4.8 Cross-analysis: community profiles

The cross-analysis shows that the detected communities are not only structural clusters but also differ in affective tone and topical focus. By joining Louvain community membership with sentiment, emotion and named entities, the analysis produces community profiles that connect interaction structure with discourse content.

This step directly addresses the research question because it shows how different parts of the conversation are associated with different emotional and thematic patterns. However, this interpretation must remain cautious because community labels are available only for authors belonging to the giant component.

Comm unity	Author s	Texts	Avg VADER	Dominant emotion	Top entity	Interpretation
C0	49	54	−0.017	Disgust	Anne Hathaway	Largest cluster; mixed and slightly critical cast-centred discussion
C2	6	13	+0.098	Anticipation	Colossal	Small anticipatory thread, mildly positive
C5	6	8	+0.093	Sadness	Anne Hathaway	Cast-focused, possibly nostalgic discussion
C6	11	19	−0.011	Trust	(DATE entity)	Neutral/trust-oriented community; top entity uninformative
C9	4	5	+0.077	Anticipation	(noisy NER)	Small anticipatory micro-community; NER unreliable at this size

Table 4. *Profile of Louvain communities in the giant component (communities with at least five texts). "DATE entity" and "noisy NER" indicate cases where the top entity is a NER artefact (e.g. the token a year classified as DATE) rather than a meaningful thematic anchor; the small SpaCy model becomes unreliable on very small communities.*

Community-level content analysis concerns only about 110 texts, corresponding to roughly 2% of the full corpus and about 2.6% of the English sub-corpus. The table should therefore be interpreted as a qualitative profile of the connected core of the discussion, not as a description of the entire corpus. Communities with fewer than five texts are not reported because their profiles are too small to support meaningful interpretation.

5. Limitations

5.1 Platform and query limitations

The analysis focuses only on Bluesky and therefore cannot be interpreted as representative of the entire audience of *The Devil Wears Prada 2*. In addition, the dataset is query-based, which means that posts discussing the film without using the selected keywords may have been missed.

5.2 Time window limitations

The dataset covers the period from 1 April to 21 May 2026. Discussions after 21 May are not included. Therefore, later reactions, long-term reception and subsequent box-office developments fall outside the scope of the analysis.

5.3 Language limitations

Sentiment, emotion and NER are computed only on the English-tagged sub-corpus. Portuguese, Spanish, French and other non-English posts are described in aggregate but are not analysed with language-specific NLP pipelines.

5.4 Network limitations

The reply network is highly fragmented, and Louvain communities are defined only on the giant component. As a result, community-level content analysis covers only a small subset of the corpus and cannot be generalised to all collected texts.

5.5 Model limitations

Lexicon-based models such as VADER, AFINN and NRCLEX can struggle with sarcasm, irony, memes and context-dependent references. In addition, SpaCy `en_core_web_sm` is a small NER model and may misclassify hashtags, usernames or short informal tokens (e.g. `ai`, `tucci` or `devilwearsprada2` sometimes classified as locations). The errors do not affect the dominant ranks but are visible on individual entries of the entity tables.

6. Conclusion

This project analysed Bluesky discussions around *The Devil Wears Prada 2* by combining social network analysis and content analysis. The results address the four sub-questions introduced in 1.3 as follows.

RQ1 — Network structure. The reply graph is highly fragmented: 1,555 authors form 371 disconnected components, and the giant component covers only 5.7% of nodes. Community detection on the giant component yields a stable modular structure: Louvain, Greedy Modularity and Girvan–Newman converge with pairwise ARI ≥ 0.96 and modularity around 0.42, while FluidC produces a coarser, divergent partition (modularity 0.289).

RQ2 — Sentiment and emotion. The English sub-corpus is polarised rather than uniformly positive (43% negative, 38% positive, 19% neutral). The dominant emotions are anticipation and trust, consistent with a release-window discussion. The Louvain communities exhibit distinct affective profiles, including clusters skewed towards joy, sadness and disgust, even though this comparison is limited to the connected core of the network.

RQ3 — Temporal dynamics. Engagement is strongly event-driven and concentrated around the release window. About 88% of the corpus belongs to the post-release period, indicating that users reacted mainly to concrete events and visible media signals rather than to a stable pre-release fandom discussion.

RQ4 — Entities and narratives. Named Entity Recognition shows that the conversation is anchored in cast members (*Anne Hathaway, Meryl Streep, Emily Blunt, Stanley Tucci*), characters (*Miranda Priestly, Andy Sachs*), fashion-industry referents (*Anna Wintour, runway*) and press-tour guests (*Lady Gaga, Doechii*). The cross-analysis with sentiment shows that the same entities recur in both positive and negative texts, with different relative frequencies depending on the affective tone.

Overall, the project demonstrates how combining reply-network analysis with sentiment analysis, emotion detection and Named Entity Recognition can reveal not only what users discuss, but also how different interaction clusters organise around cultural events and assign meaning to them. The fragmented structure of the Bluesky conversation about this topic is itself a substantive finding: on this platform, the discussion takes the shape of a constellation of micro-reactions to a small number of viral seed posts rather than that of a single sustained debate.